

DATUM LINE

AI × BIO · 9 DEC

“Do my cells look normal?”

Confidence with controls.

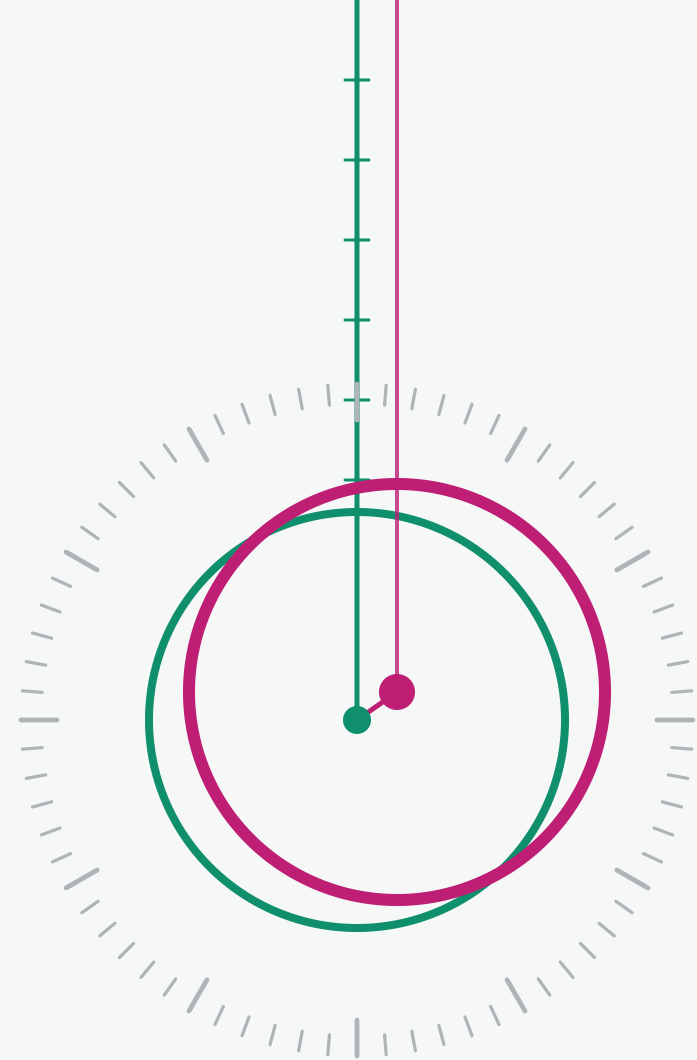


control image

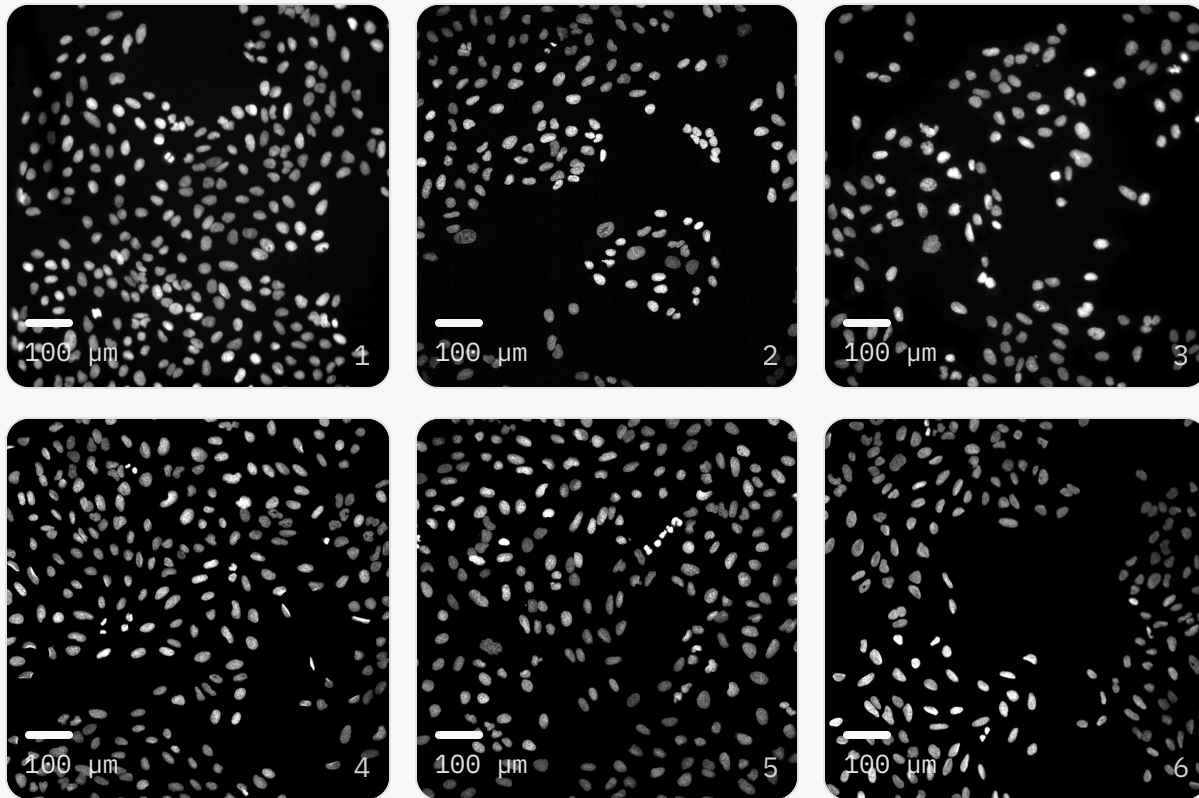
no compound, no result

ask

Kat Titterton · San Francisco



One lab, or six?



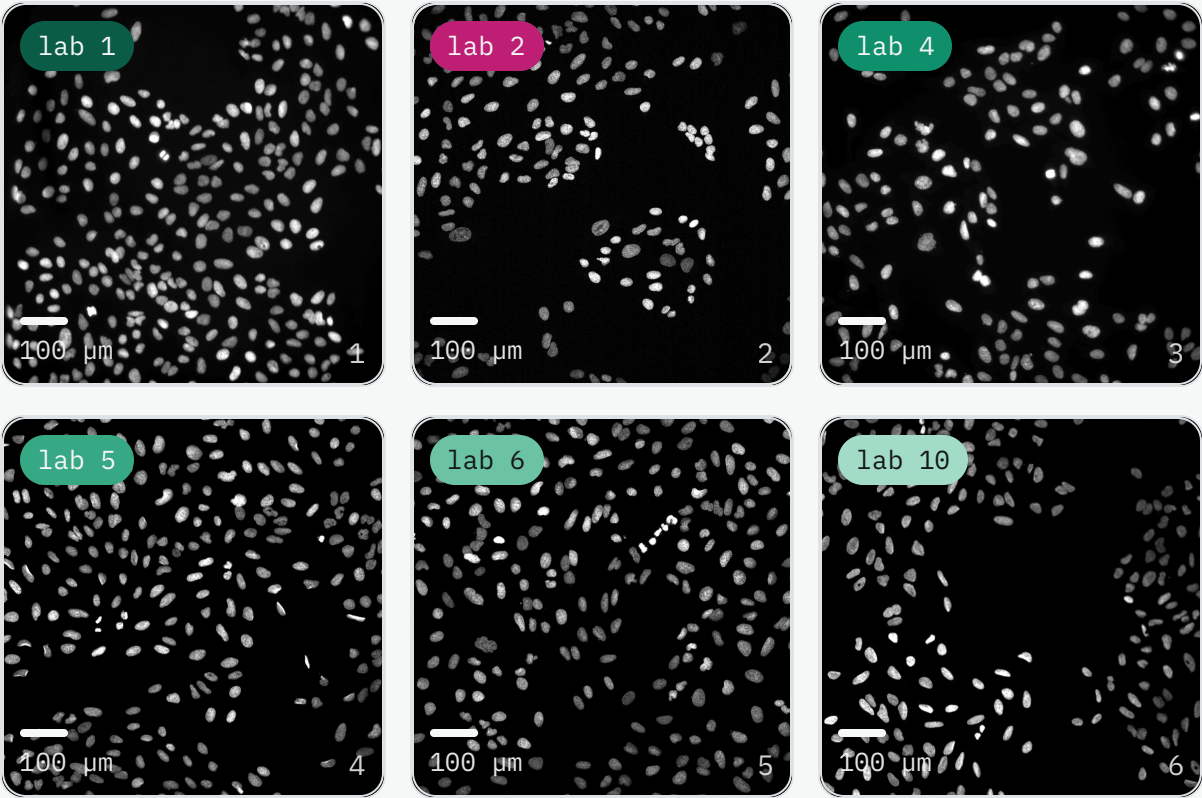
Cell Painting five dyes, 3,773 numbers per well

-  same line ✓
-  same protocol ✓
-  no drug ✓

Fig. 1 93,228 untreated wells, 11 labs, plates held out. JUMP [7]

[click → the answer](#)

Could you tell? A classifier can.



Cell Painting five dyes, 3,773 numbers per well

99.9% CLASSIFIER

9% CHANCE, 1 IN 11 LABS

78 wells wrong, of 93,228

99.9% → 24% normalised to itself. A3

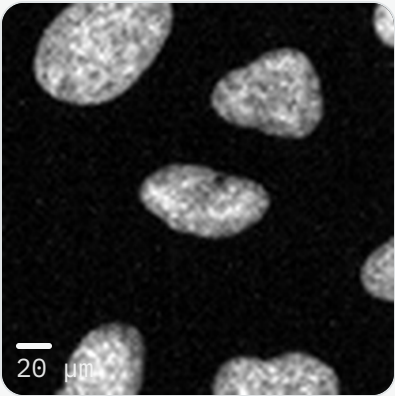
Fig. 1 93,228 untreated wells, 11 labs, plates held out. JUMP [7]

click → follow one well



One well, **all the way out.**

ONE NUCLEUS



192 μm^2 · 0.59

ITS WELL

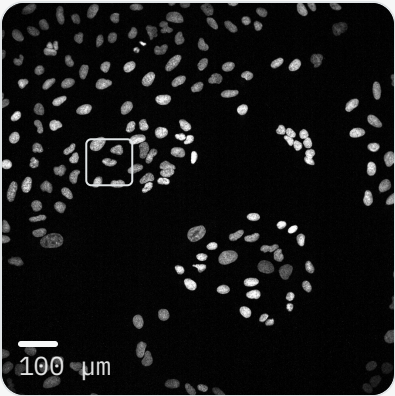


plate 1053600681

3,773
numbers
→
one point

WHERE IT LANDS

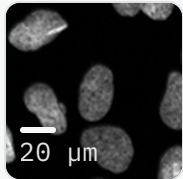
Cell Painting five dyes, 3,773 numbers per well



93,228 wells, 11 labs

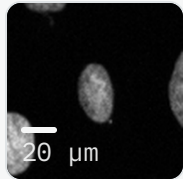
other labs

lab 5



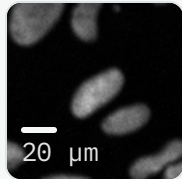
195 μm^2 · 0.94

lab 6



213 μm^2 · 0.50

lab 10



207 μm^2 · 0.29

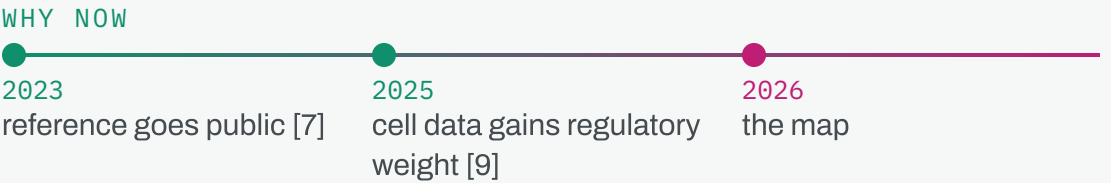
AREA $\approx \approx$
TEXTURE $\neq \neq$

Shape, not count.

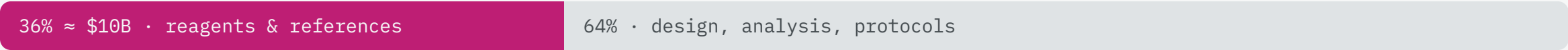
Fig. 1 93,228 untreated wells, 11 labs, plates held out. JUMP [7]

click → start over

Today: search, ask, guess.



IRREPRODUCIBLE PRECLINICAL RESEARCH · US · PER YEAR [6] \$28B



1

search

A grid of six microscopy images showing U2OS cells. The images display different staining patterns: some with orange/red nuclei, others with blue nuclei, and some with green or white signals.

2

ask around

To: whoever ran this line last

Subject: does this look right to you?

Attached: screenshot.png

3

guess

\$ RUN IT ANYWAY · TOO FAST

hundreds of thousands

of R&D

⌚ HOLD · TOO SLOW

months

of development

[6] Freedman et al., *PLoS Biol*, 2015. Six real U2OS papers.



macrofluidic.me/resume

TEAM

One founder today.
Looking for a technical co-founder.
Collaborating labs: insitro, Axiom Bio, Soley.

Katherine Titterton

Five labs. One **unanswered question**.

●

2023–2026

Axiom

SAN FRANCISCO

founding MTS • hepatocytes at scale

●

2022–2023

insitro

SOUTH SAN FRANCISCO

neuroscience assays • productionization

●

2021–2022

MIT

SYNTHETIC NEUROBIOLOGY

expansion microscopy

●

2018–2021

abbvie

NEUROSCIENCE

tau • neurons

●

2017–2018



HARVARD MEDICAL SCHOOL

Xenopus oocytes • GABA_A receptors

●

2017

COLGATE

BA

neuroscience + math biology

MY WORK

- 1

Cell Painting in primary human hepatocytes

Cell Systems 2026 • Axiom Bio
- 2

Counting cells predicts bioactivity benchmarks

Nature Communications 2026 • Axiom Bio
- 3

A neuronal tau aggregation assay

J Biol Chem 2025 • AbbVie
- 4

Multiplexed expansion revealing

Nature Communications 2024 • MIT
- 5

Neurosteroid photolabeling of GABA_A receptors

Eur J Med Chem 2019 • MGH • Harvard Medical School

PUBLISHED WITH

Broad Institute • MIT • AbbVie • EMBL-EBI • HESI • Bayer
• University of Cambridge • Uppsala University

Eleven labs. Eleven normals.

6.2

spreads apart

99.9%

told apart

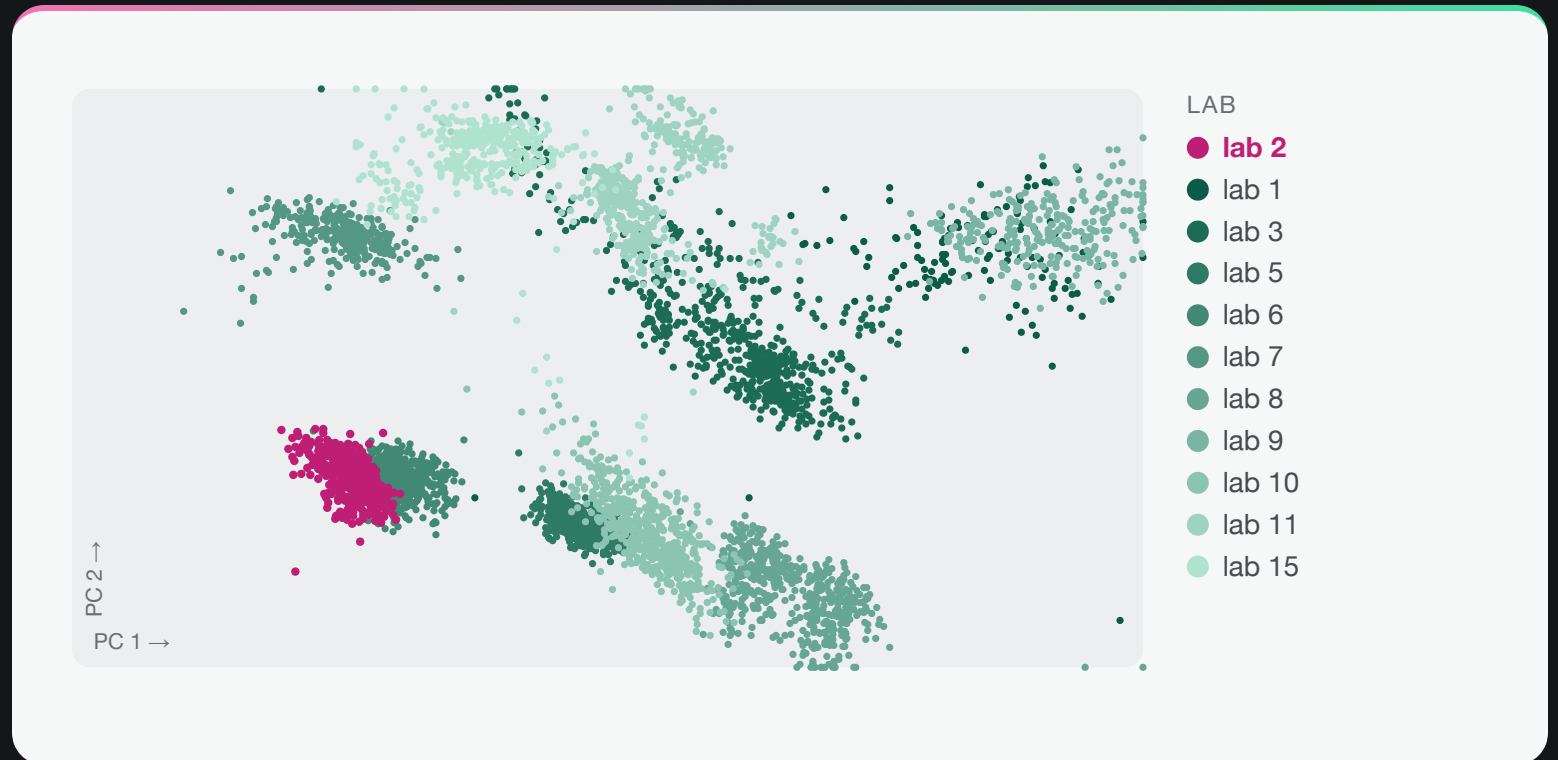
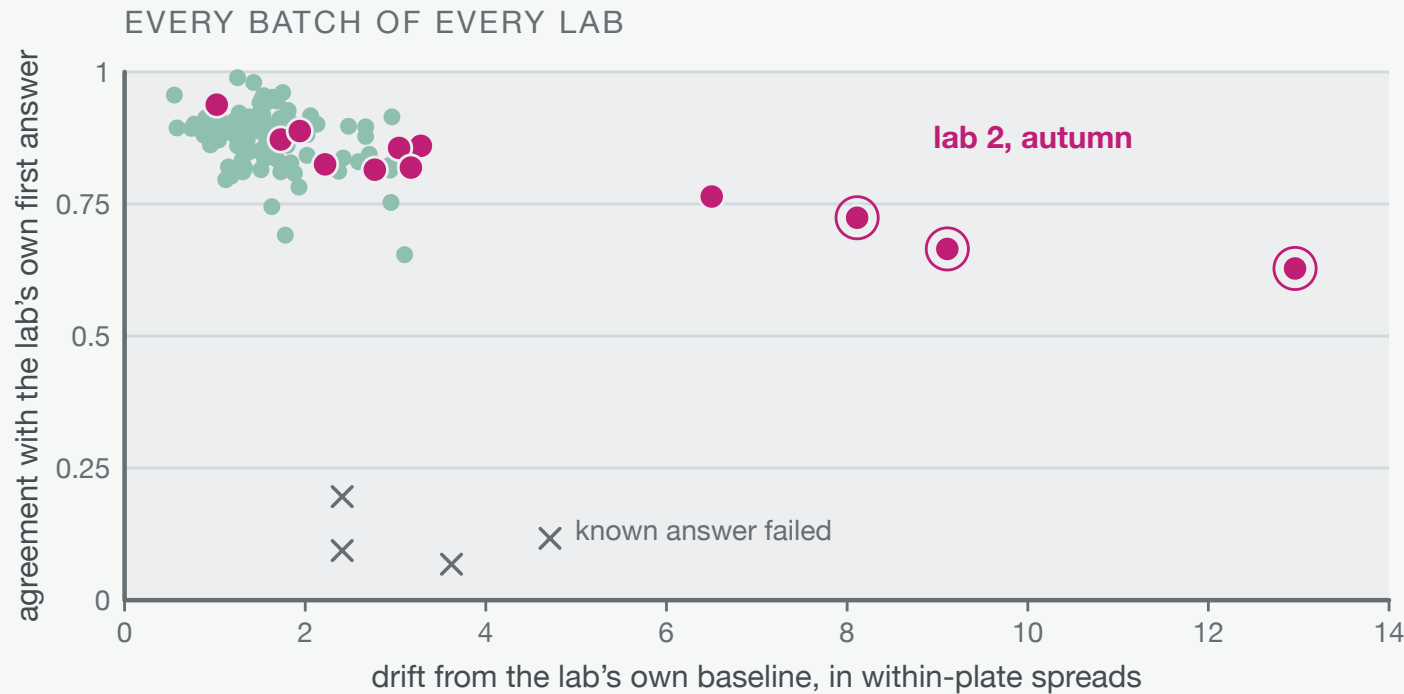


Fig. 2 15,352 untreated wells, two of 3,773 numbers shown. JUMP [7]

[click → why the map matters](#)

Where you sit predicts what you measure.



INSIDE LABS

$\rho = -0.71$
drift, against your own answers

lab 2, same drugs

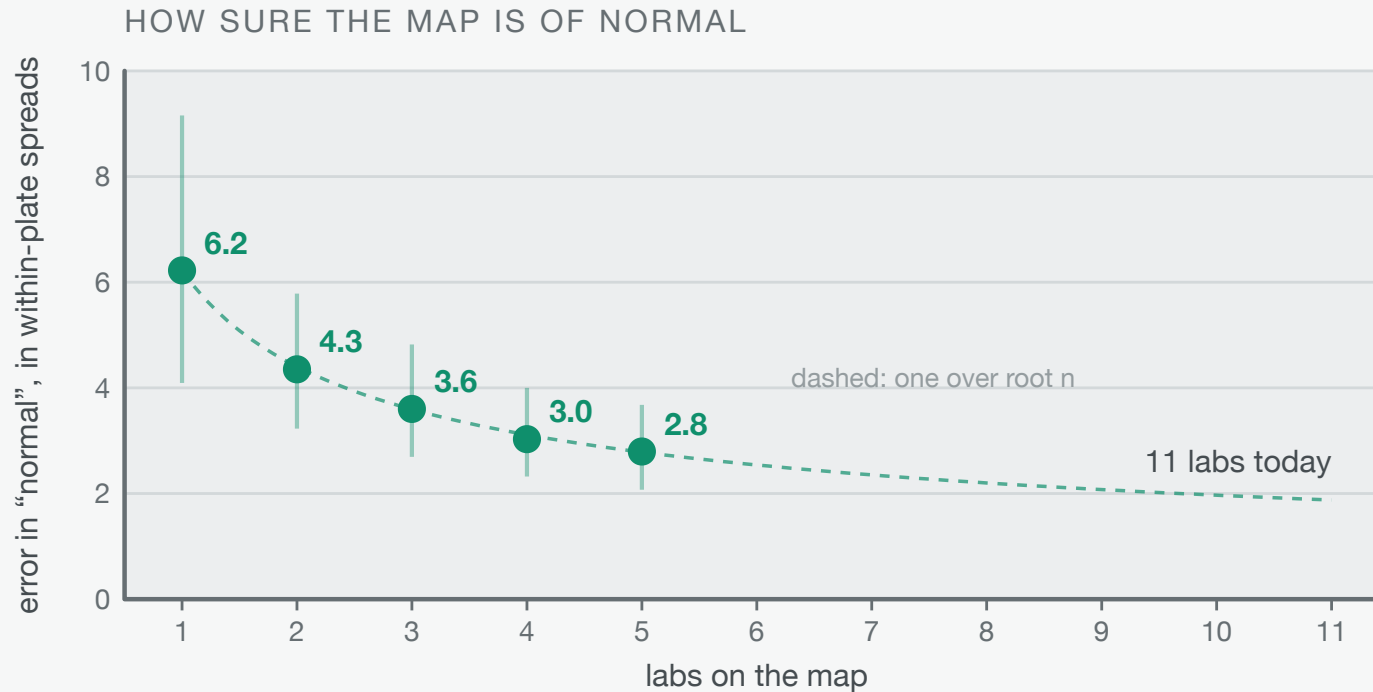
0.85 before

0.67 after

Fig. 3 one point per batch, 118 batches. JUMP [7]. A13

click → start over

Every lab that joins **sharpens the map.**



THE MAP'S ERROR

6.2 → 2.8

error in normal, one lab → five

your own baseline

0.9 → 0.4

one plate → eight

 COMMUNITY [open](#)

 DATA ACCESS [paid](#)

 THE BENCHMARK [co-built](#)

Fig. 4 two maps from different labs, how far apart. 1,871 plates. JUMP [7]. A14

Drop an image. **Get an answer.**



lab 2 · plate 1086292259 · 16 August 2021
one control well, DNA

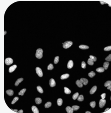
answered



PC 2 →
PC 1 →

you

4.4 to lab 6



Yours, and lab 6's. 4.4 apart.



The answer

LAB 2 · 16 AUGUST 2021



YOUR CELLS
U2OS · 25 plates

WHERE YOU SIT
in distribution

2.8 from your first batch · 4.4 to lab 6, the nearest

YOUR OWN CHECK
188 cells per well · in range

THE MAP BEHIND IT
1,652 plates · 10 labs

YOUR DATA, MADE COMPARABLE
agreement with other labs **0.70 → 0.77**

Fig. 5 lab 2, 16 Aug, replayed. JUMP [7]. A11

[click → the leaderboard](#)

Fig. 6 score: 100 minus mean percentile, untreated wells. JUMP [7]. A17



LAB 2 · 16 AUG

YOUR BEST
#48, 2 Aug · score 58

TO CLIMB
steady: 2.9 against 1.2 for the field

THE LAB TO BEAT
lab 5 · median score 70 over 33 batches

```
click → start over
```

Five questions, five tools.

Use all of them.
One did not exist.

TOOL	THE QUESTION IT ANSWERS	WHAT IT DOES NOT DO
STR authentication	Is it the line the label says?	nothing about behaviour
vendor certificate	Was it fine when it shipped?	nothing after shipping
plate controls	Did this plate work?	cannot see a whole lab drift
phenomics platforms	Is my own data consistent?	removes lab differences by design
Datum Line	Where do my cells sit, against every other lab?	cannot say which line; fluorescence today

Eighteen months, four milestones.

\$1.8M pre-seed · A21

1



three labs on the map

insitro · Axiom Bio · Soley

2



**model that reads any
image**

99.6% today · A19

3



**benchmark, with two
partners**

A16

4

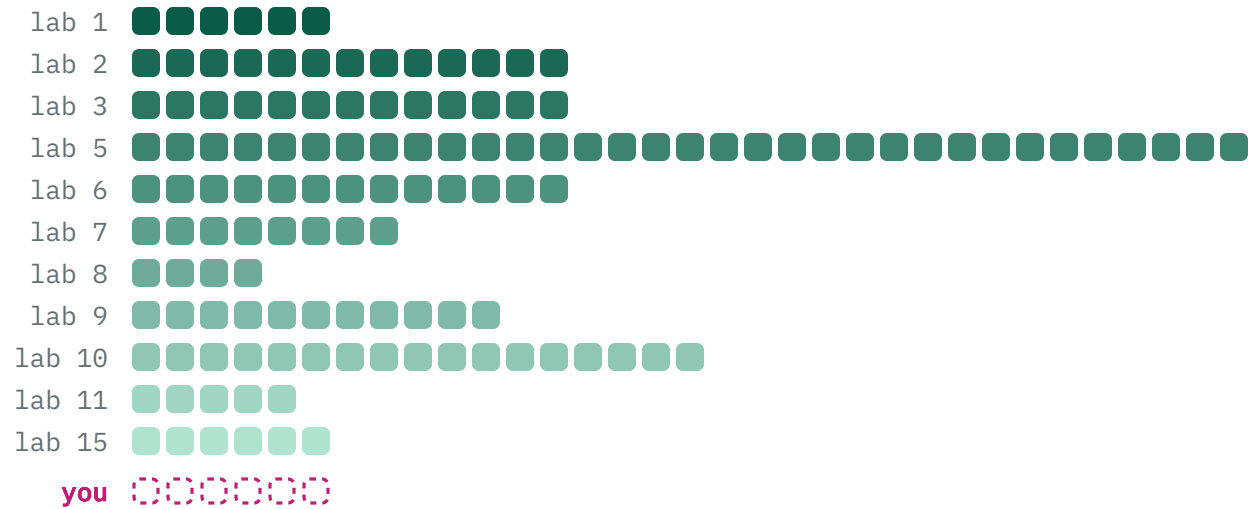


first paid certificate

Every plate you image, placed on the map.

One map, for people and agents.

FIG. 7 • ALREADY ON THE MAP 129 batches • 11 labs • JUMP [7]



1,871 plates. Next row is yours.

WHAT I NEED

 images

 people

 bench

 pre-seed

 partners

Have control images? Talk to me.

The rest, if you ask.

Find the question, jump to the board. Every number traces to a file in results/ or a numbered reference; pending work says pending.

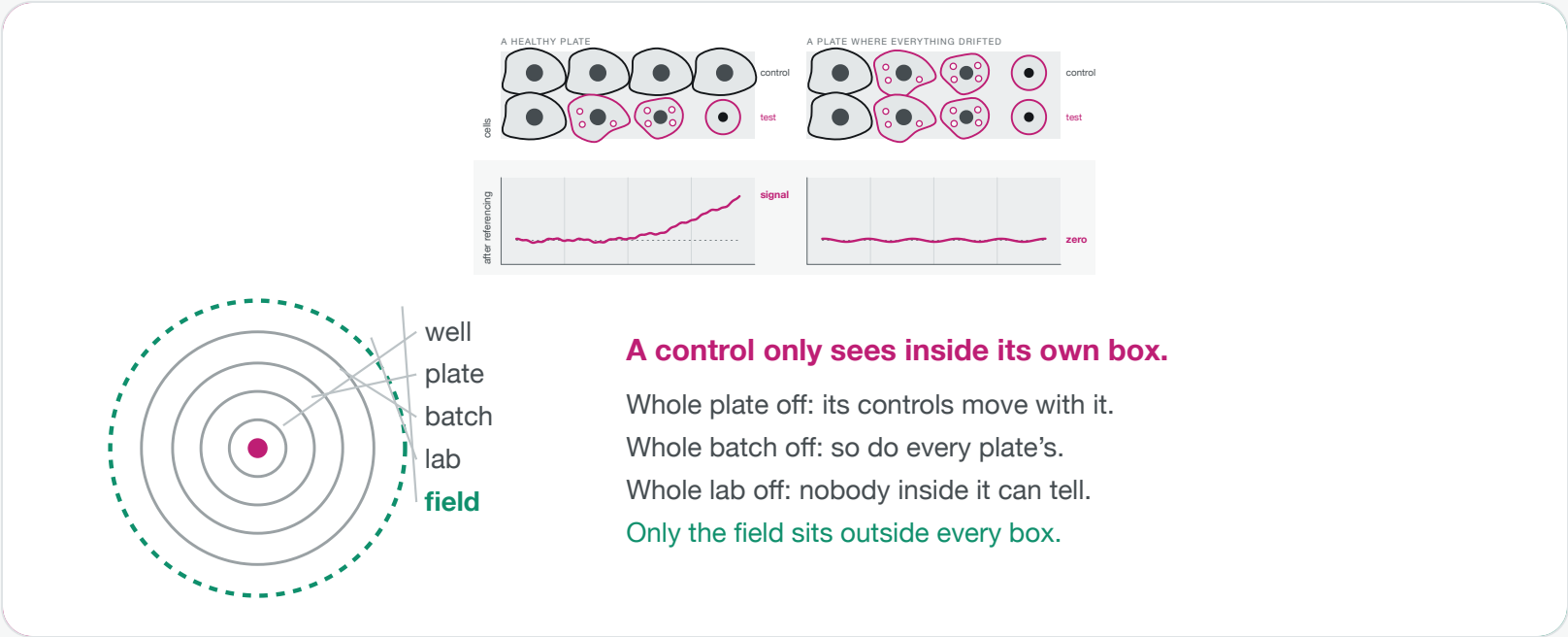
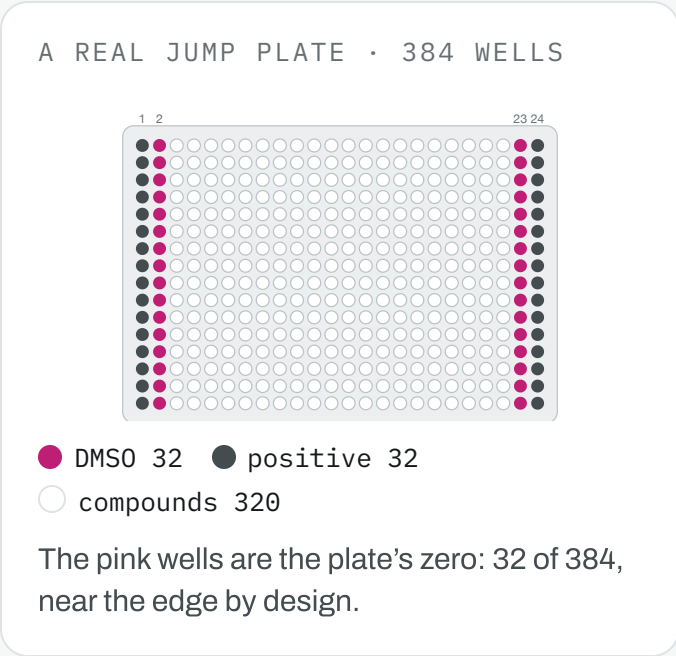
A11	Would it cry wolf? Six of 129 batches
A12	What moved? Stain intensity, not cell count
A13	How does one lab move over time? The lab's check and the field's view
A14	What does the field's frame buy? Agreement between labs

A15	Who pays? Three buyers, one certificate
A16	What would the money build? The ugliest dataset, on purpose
A17	Is the leaderboard fair? Controls only, never results
A18	What is private? Control wells leave; nothing else does

A19	What is the machine learning? Image in; place, distance, move out
A20	What about biosecurity? Cells cannot lie about where they came from
A21	How much, and what for? Planned at \$0.9M, asked at 2x
A22	How big? Bottom up, every assumption on the board

There is no control well.

Every result is test minus control, on the same plate. But a whole plate can be off, or a whole batch, or a whole lab. A control only sees inside its own box.



$$x_{test} = \mu + s + \tau(s) \quad x_{ctrl} = \mu + s \quad \Delta = \tau(s)$$

s , the plate's state, cancels from the number. Never from the cells.

So a control only sees inside its own box. If the plate is sick, so is the control. Then how would anyone know?

Fig. A1 Where a control sits, and what it can see. Left, a real 384-well JUMP compound plate (lab 2, 1053601756): 32 DMSO, 32 positive, 320 compound wells. Right, schematic cells and traces, and the nesting of well, plate, batch, lab and field. No measurements shown.

Six operations on one embedding.

Every well becomes a vector $e \in \mathbb{R}^{3773}$, z-scored across the field. Everything on the previous slides is one of these, and each one is a difference of means or a held-out classifier, so a lab can check it.

1 · EMBEDDING

[1]

$$e = \varphi(\text{image}); \varphi \in \{\text{CellProfiler}, \text{CNN}, \text{ViT}\}$$

Which φ barely matters: the three performed similarly on hepatocytes. The extractor is not the asset.

2 · LAB CLASSIFIER

99.9%

$$f: e \rightarrow \text{lab}, \text{ held out by plate}$$

Logistic regression, L2, five folds. Balanced accuracy, chance 1/11. Permuted labels 9.3%. Without the 1536-well labs: 100.0%.

3 · DISTANCE, THE CERTIFICATE

[e6]

$$d = \| E[e \mid \text{plate}] - E[e \mid \text{field}, c] \|_{\sigma}$$

Difference of two means in units of the field's spread, conditioned on c . Its direction is what moved; $-d$ is the way back.

4 · VARIANCE DECOMPOSITION

[e5]

$$\text{Var}(e) = V_{\text{model}} + V_{\text{lab}} + V_{\text{batch}} + V_{\text{plate}} + V_{\text{well}}$$

Nested, so shares sum to one. Microscope model separates from lab because models are shared across labs.

5 · REGRESS VARIABLES OUT

24%

$$e_{\perp} = e - \hat{E}[e \mid c], \text{ then re-run } f$$

Remove only what you know is instrument, keep what may be biology. Remove each plate's own control mean and f falls to 24%.

6 · MORPHOLOGY CHANGE, TWO WAYS

 δ

$$\begin{aligned} \text{DISTANCE } \delta &= f(\|e - E[e \mid \text{vehicle}]\|), f \text{ bounded} \\ \text{CLASSIFIER } g: e' &\rightarrow \text{label}, e' \text{ aligned to the reference before comparison} \end{aligned}$$

Same vector, two questions. Discovery wants invariance to the lab: align first. QC wants the lab coordinate: never align it away.

results/e1, e1b, e5, e6 · [1] Ewald et al. 2026

What JUMP records. What nobody does.

Normal is a conditional expectation, $E[x \text{ / every variable below}]$. Even JUMP, built to be comparable, records only the instrument side.

RECORDED IN JUMP

- lab
- batch date
- plate
- plate type
- well position
- microscope
- confocal or widefield
- laser or LED
- objective NA
- fields per well
- filter set
- pixel size
- brightfield planes
- CellProfiler version

NOT IN ITS PUBLIC METADATA

- DMSO % per well
- DMSO lot
- DMSO water uptake
- stock freeze-thaw
- serum lot
- media lot
- passage
- thaw date
- seeding density
- confluence at dosing
- dosing to fixation
- stain and dye lots
- operator
- incubator shelf
- edge evaporation
- mycoplasma status
- CO2 and humidity

NORMALISE THE LAB AWAY AND THE LAB DISAPPEARS

- 99.9% raw features
balanced accuracy naming the lab
 - 24% plate-centred
own control mean removed; chance 9%
 - 19% plate-centred and scaled
own control spread removed too
- 29,943 wells, 16 per plate, plates held out. results/e1-redteam.

Centre each plate on its own controls, the standard first step, and the lab is named 24% of the time. It removes most of the coordinate a certificate needs.

Batch correction is built to remove this signal: Arevalo J, et al., Nature Communications, 2024. Variables: JUMP datasets metadata.

The benchmarks are so weak that counting cells often wins.

A paper I coauthored.

Accurately predicting the activity of a chemical in each bioactivity assay based on its already-known properties is extremely useful in drug development. Unfortunately, we discovered that many assays in widely used assay-activity benchmark datasets directly relate to cell health and cytotoxicity. Many other assays intend to capture a more specific phenotype, but their active compounds impact cell count, while inactives do not. In both cases, counting cells achieves unexpectedly high performance in these benchmarks, making them less useful for discerning whether additional properties, such as phenotypic profiles (mRNA or Cell Painting), provide additional useful information on bioactivity. To accomplish this goal, we recommend filtering benchmarks to exclude such assays and including a cell-count baseline. Using a benchmark with 24 protein-target assays, we confirm that models leveraging Cell Painting image-based profiles outperformed the baseline cell count model. We propose several other practical recommendations for benchmarking machine learning models for predicting bioactivity and assessing the added value of mRNA, protein, or image-based profiles.

ABSTRACT, VERBATIM

the finding

the recommendation, and what held up

209 and 270

assays in the benchmarks the field scored itself on; prior work reported AUC above 0.9 on 32% of 209

24

protein-target assays left once cell-health assays were filtered out; there, profiles beat the count

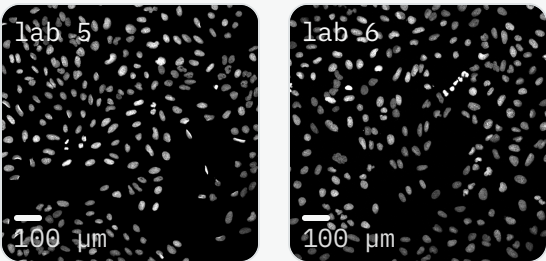
WHY IT MATTERS HERE

Whoever sets the baseline sets the benchmark. That is why the benchmark is a milestone, not a paper: planted failures, a cell-count baseline on every task, every model graded (A16).

Two ways to ask the same question.

Why now: an image classifier can answer this, and until recently nothing could. A count moves only when cells die or divide; an embedding of the whole well moves when anything changes.

SAME CELL LINE, SAME PROTOCOL



cell count	236	224	same
shape	0.94	0.50	different

METHOD ONE

Does this run belong with the past runs?

$$g: e' \rightarrow \text{belongs} / \text{does not}, \quad e' = \text{align}(e)$$

Align first, so the answer is the cells and not the day.

Standard alignment; the tuning is ours.

METHOD TWO

How far out, and which way back?

$$\delta = f(\|e - E[e \mid \text{vehicle}]\|), \quad f \text{ bounded}$$

Bounded, so one wild well cannot swamp the score.

Distance published with the Broad [1]; transform unpublished.

Both methods need the same thing to work: somebody else’s images. It takes a reference to know which.

Fig. A5 Same cell line, two untreated wells, and the two readouts. Left, DNA channel, JUMP cpg0016, CC0; nuclei by Otsu segmentation; detail = high-frequency energy over total, inside each nucleus. Right, the two readouts: a classifier on aligned embeddings, and a bounded distance from the vehicle centre. Alignment and distance follow published practice [1]; the specific transform and tuning are not disclosed here. [1] Ewald JD, Titterton KL, et al. *Cell Systems*, 2026.

The line is not expired. Your baseline is.

Two plates of one lab sit further apart the longer the gap between them. In lab 2 the distance is still rising after thirteen weeks.



IS THE CELL LINE RUINED?
Not known from profiles alone. The data says the lab moved and when. It does not say why.

SO WHY RE-CERTIFY?
Because your certificate is a comparison made on one day, and the baseline it was made against goes stale within weeks.

HOW SOLID IS THE NUMBER?
One lab, 219 plates, 13 batches, June to October 2021. 0.63 in the same batch, 4.7 over 13 weeks. Other labs are not yet measured.

STR says who. Images say how.

Not a competitor: the other half of the same certificate.

STR AUTHENTICATION



17

loci

once

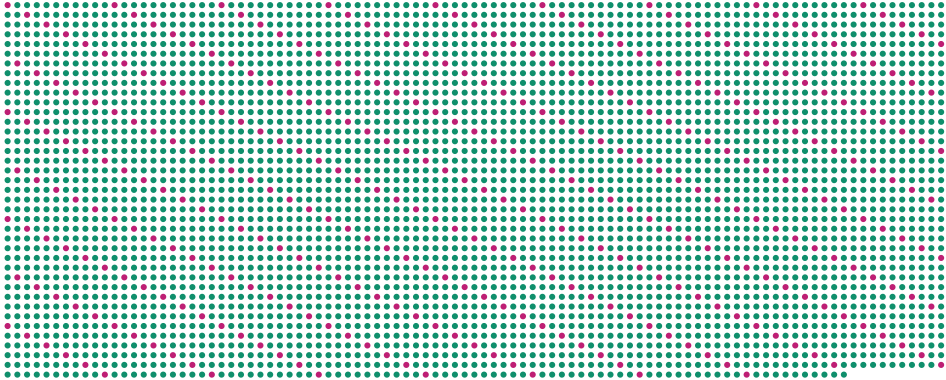
per line, from
a mailed card

who

which line it
is

The standard for identity. Two cultures with the same fingerprint can behave differently.

DATUM LINE



3,773

numbers per well

every batch

from images already taken

how

the way it behaves, against
every lab

Behaviour, placed on the field's map. Cannot say which line it is.

identity · STR

+

behaviour · Datum Line

=



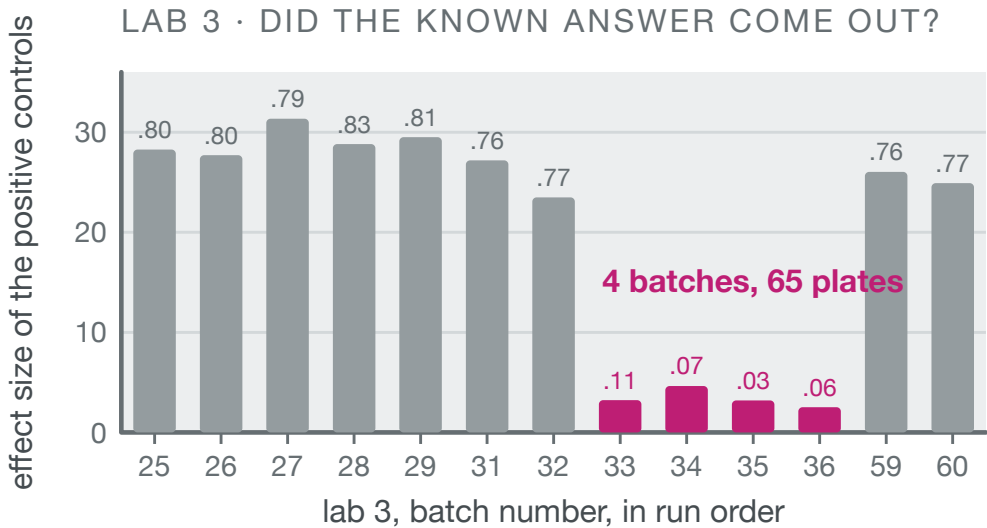
one certificate.

Partner, not rival: the STR provider is the natural first integration.

STR: ATCC cell authentication service, 17 loci plus amelogenin. Datum Line: 93,228 untreated wells, 1,872 plates, 11 labs, JUMP [7].

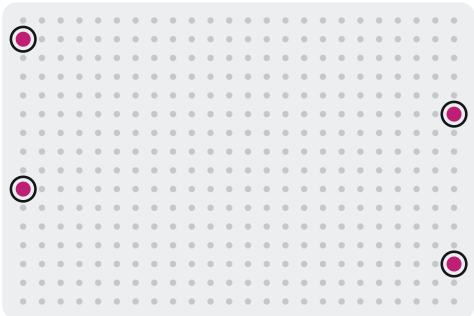
A known answer, run in every lab.

Untreated wells certify where a lab sits.



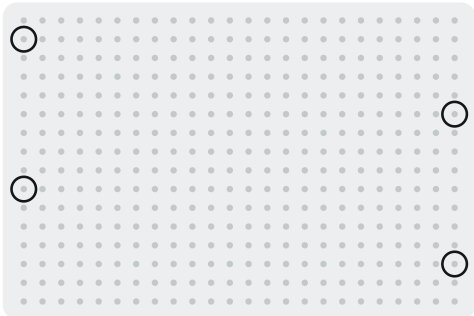
WHERE ON THE PLATE DOES AMG900 SHOW?

a batch that came out
plate C13443aW



○ where the plate map puts it

a batch that failed
plate A12069aW



● wells that look like it

RUNS NOW ON JUMP

Every compound plate carries eight positive controls. In 125 of 129 batches the known answer came out.

WHAT THE FIELD CAUGHT

Lab 3, 4 consecutive batches, 65 plates: the wells the plate map labels as positive controls show no effect, and no other well on the plate does.

TWO CHECKS, TWO FAILURES

The untreated-well certificate flagged 1 of these 4 batches. A plate can sit in the right place and still not answer.

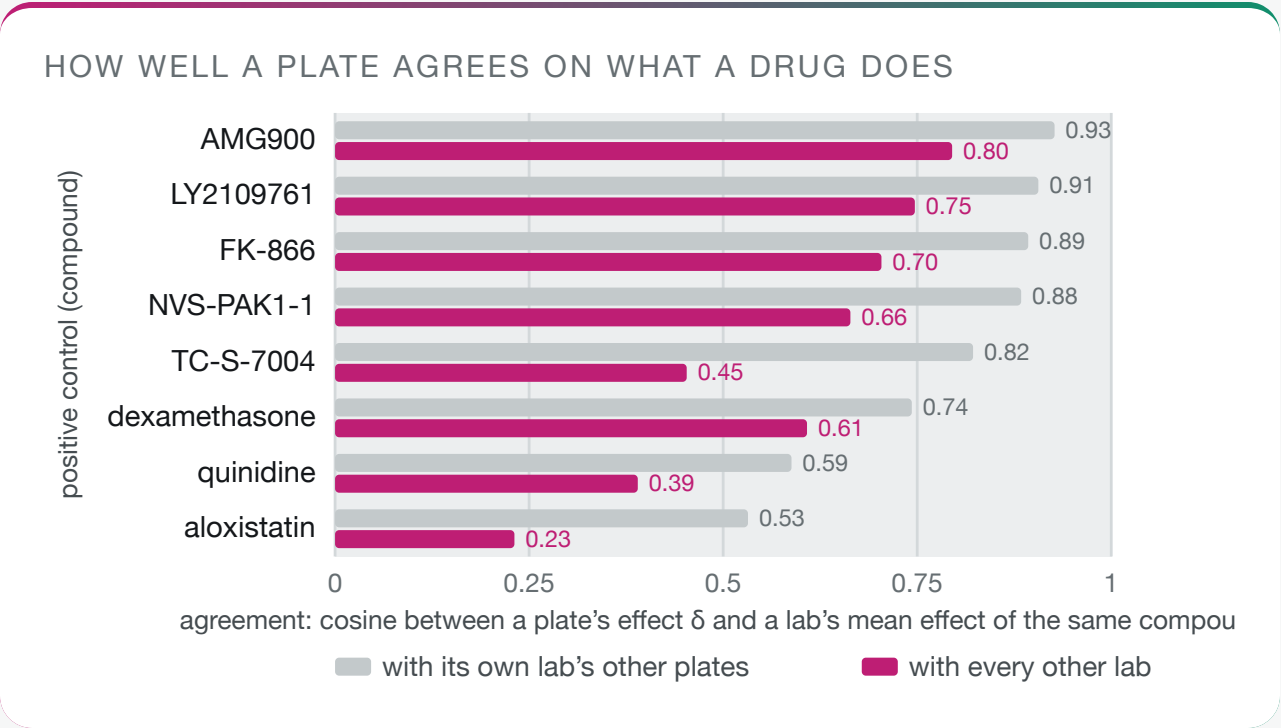
Same drug. Different answer.

Labs running the same assay slightly differently rank the same compounds differently.

WHAT A SCREEN REPORTS, PER PLATE

$$\delta_{p,c} = E[x \mid \text{plate } p, \text{ compound } c] - E[x \mid \text{plate } p, \text{ DMSO}]$$

Treated minus control, on the same plate. Scaling fit on the field's DMSO wells only, never on the plate being judged.



Q1 · DOES IT AGREE? [E8]

Less with other labs, for every compound.
Grey: a plate's δ against the mean δ of its own lab's other plates. Pink: against each other lab's mean, averaged. AMG900: 0.93 and 0.80.

Q2 · DOES IT TRACK THE CONTROLS? [E8]

Inside labs, yes.
Spearman between a plate's control-well distance to the other labs and its disagreement with them: 0.20 to 0.53 by compound; pooled 0.30, shuffled 0.01.

Q3 · DOES IT MATTER? [E8]

Effect size differs 14% to 79% between labs.
Effect size is $|\delta|$. Coefficient of variation of the lab means, by compound; inside one lab, plate to plate: 10% to 44%.

Shown in fluorescence.

Bet on brightfield.

Every number here is Cell Painting: five stains. The product reads brightfield: none. That gap is the bet, testable on public data.

WHAT IS SHOWN

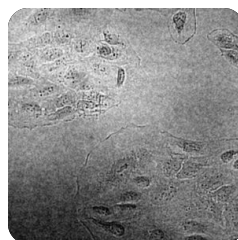
Stained wells name the lab

Untreated U2OS wells, 11 labs: the lab is named 99.9% of the time, chance 9%.
results/e1

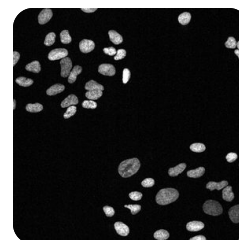
WHY THE BET IS FAIR

Fewer variables, not more

No stain lot, dye lot or staining step: three unrecorded variables gone (A3). JUMP images brightfield beside the stains, so the same wells re-run without them.



BRIGHTFIELD



DNA STAIN

One field of one well, two ways.
Lab 2, June 2021.

WHAT WOULD KILL IT

Brightfield alone at chance

Then the certificate asks for one cheap stain. First test: a frozen vision model names the lab from one brightfield field at 99.6%, 8 labs (results/e15). **Next: the whole map from brightfield.**

Brightfield timecourse is the modality I build screens on.

Would it cry wolf? 6 of 129.

A flag is only worth something if it is rare, and if it survives the choices made to compute it.

LAB 2 · THE SAME 13 BATCHES, JUDGED TEN WAYS

	7	14	21	12	19	26	2	8	16	23	30	20	3	
	Jun	Jun	Jun	Jul	Jul	Jul	Aug	Aug	Aug	Aug	Aug	Sep	Oct	
as published	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
baseline is the first two batches	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
baseline is the first three batches	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
no clipping	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
no image-level features	■	■	■	■	■	■	■	■	■	■	■	■	■	2,904 features
image-level features only	■	■	■	■	■	■	■	■	■	■	■	■	■	869 features
all-vehicle plates removed	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
lab 1 removed from the field	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
lab 8 removed from the field	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features
labs 1 and 8 both removed	■	■	■	■	■	■	■	■	■	■	■	■	■	3,773 features

HOW RARE

6 of 129 batches are outside, in 3 of 11 labs. 8 labs never flag.

DOES EVERY PLATE AGREE

In lab 2's flagged batches 25 of 25 plates are outside. In its other 10 batches, 2 of 194.

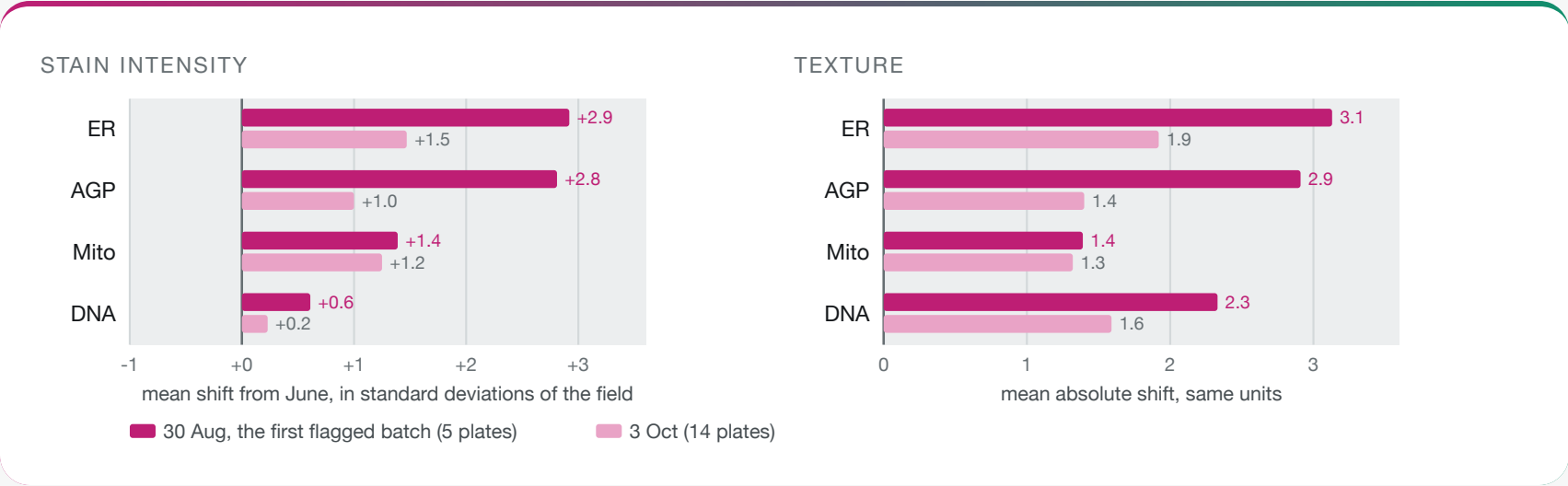
WHAT CHANGES IT

1 of 10 variants: a baseline that takes in 21 June, the batch whose cell count fell. A baseline should be a batch the lab trusts.

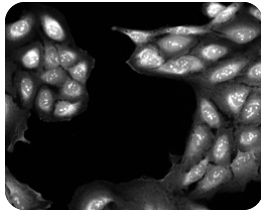
Outside: further from its own first batch than from the nearest other lab. Flagged batches hold 5, 6, 14 plates.

Intensity rose. Cell count did not.

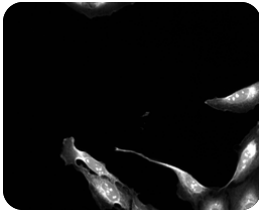
The certificate's distance is a difference of two means, so it reads feature by feature.



AS THE MICROSCOPE SHOWS THEM



7 JUN · ER
7,612 counts



30 AUG · ER
51,230 counts

Both look fine auto-scaled. Counts 7× apart; DNA 20×.
Pattern moved too: perinuclear, then whole cell.

READING

On 30 Aug, ER and AGP intensity sit 2.9 and 2.8 sd above June, and 96% or more of those features moved the same way. By 3 Oct: 1.5 and 1.0. Cells per well: 189, then 165 and 182.

INSTRUMENT OR CELLS

Image-level features carry 20% of the move and are 23% of the features, so the move is not confined to acquisition. Profiles cannot tell a stain lot from biology.

WHAT A LAB GETS

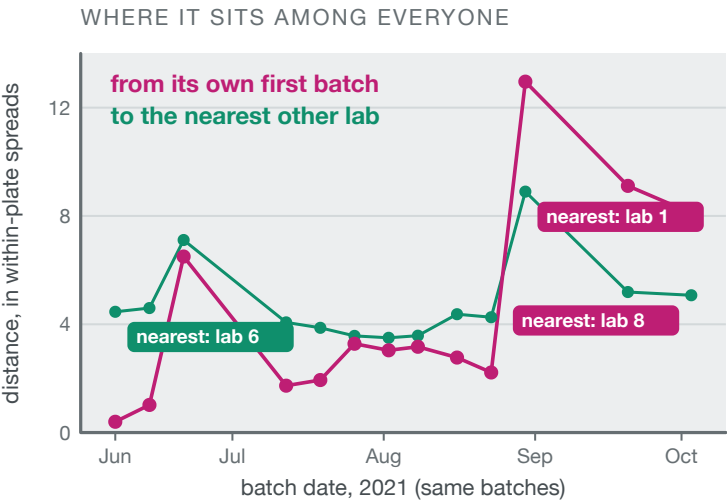
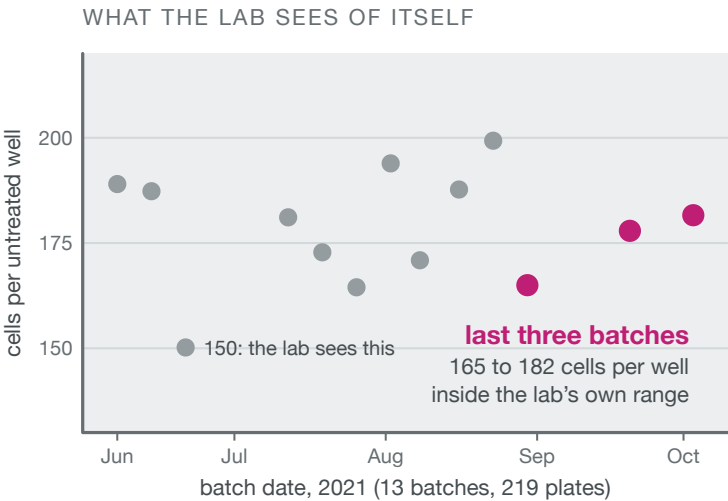
Where to look first: exposure and the ER stain, whose pattern changed as well as its brightness. Stain mix-up, bleed-through or biology are the three things to check. Not why: that is the lab's to find, a day after the run and not four months.

Shift from the lab's first batch, in standard deviations of the field; means over the features of one channel.

Alone, it looks fine.

Among everyone, it moves.

Same 13 batches, two views.



LOOKING AT ITSELF

Cells per well stay between 150 and 199.
Every batch passes. Odd things can sit
comfortably inside a lab's own range.

LOOKING AT THE MAP

From 30 Aug the batches sit further from the lab's own
first batch than from another lab. The lab they most
resemble changes: lab 6 → lab 1 → lab 8.

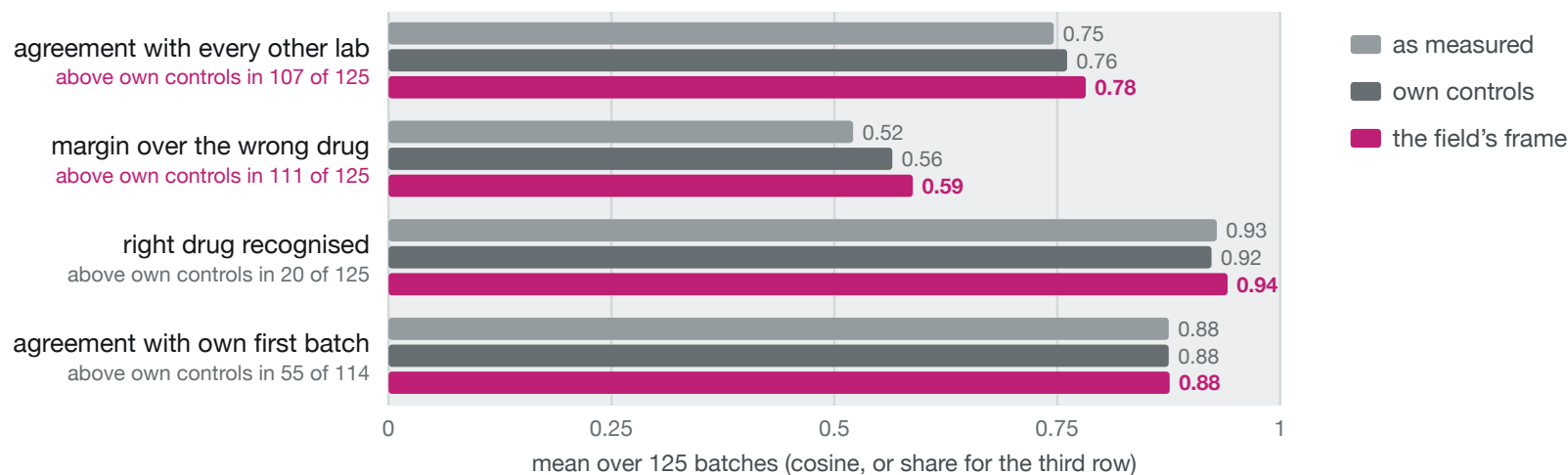
WHAT A CHANGE MEANS

Not a verdict. Resembling a different lab than
you did is a reason to look: interest when you
meant to change something, concern when
you did not.

Distances in units of the field's within-plate spread; the nearest lab is the lab whose centre is closest. Lab 2, U20S, 2021. JUMP [7]. What moved: A12.

One frame: **what it buys.**

Each batch's untreated wells are moved onto the field's, built without that lab.



DOES IT HOLD

With 100, 200 or 400 components and shrinkage from 0.05 to 0.5: above own controls in 89 to 116 of 125 batches for agreement, 52 to 123 for margin.

WHAT IT DOES NOT DO

A lab's agreement with its own first batch does not change: 0.88, then 0.88. Effect sizes vary more between batches: 0.06 becomes 0.12. The gain is between labs, and it is modest: 0.76 to 0.78 over own controls.

HOW DENSE

A lab's own baseline settles by eight plates: 0.89 to 0.39 spreads. The field's reference moves 0.41 when one lab leaves.

Three buyers, one certificate.

Contributors use it free. The labs that cannot afford to be wrong pay.

	 ACADEMIC LABS AND CORES	 PHARMA AND CRO SCREENING	 CELL BANKS AND VENDORS
what is at stake	the paper	the screen · ~\$2.6M	the lot
what they pay	free, for contributing controls	~\$5–15k per line, per quarter 170× less than one unchecked screen	a fee per certified lot
what it builds	the reference	the revenue	the network

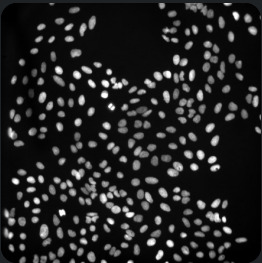
A subscription is participation in the shared data layer, membership of the community, and a very narrow window: your control wells go in, your readings come out, and nothing else is visible to anyone. Each reading is the certificate. Prices proposed.

Build the ugliest dataset in biology, **on purpose.**

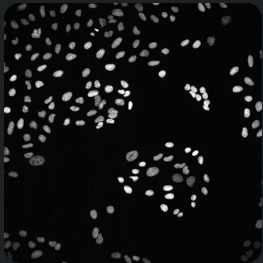
Same cells, same protocol, no drug. It already looks like this.

THE TEST SET · WHAT SIX LABS ALREADY SEND

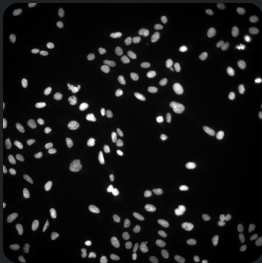
real wells, JUMP cpg0016, CC0



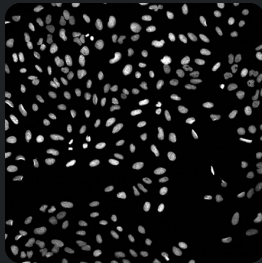
LAB 1
Phenix · widefield



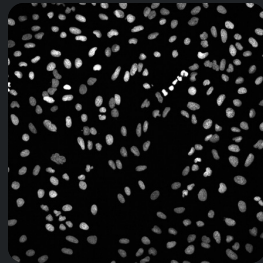
LAB 2
CV8000 · confocal



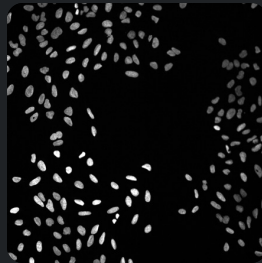
LAB 4
Phenix · widefield



LAB 5
CV8000 · confocal



LAB 6
CV8000 · confocal



LAB 10
CV8000 · confocal

5
microscope models

0.4–1.0
objective NA

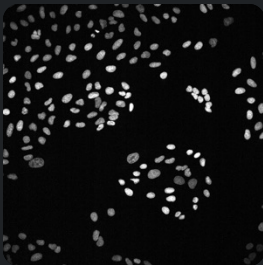
0–3
brightfield planes

0.60–0.70
µm per pixel

8
plate types

Across the 13 JUMP labs.

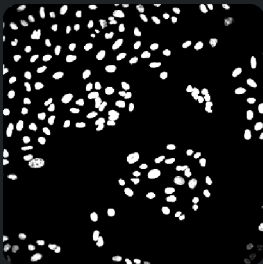
RED TEAM one well, simulated



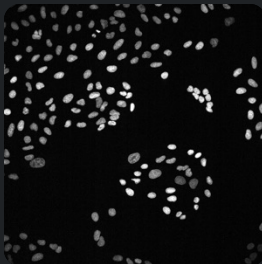
AS IMAGED



DEFOCUSED



OVER-EXPOSED



CLIPPED WELL

A public benchmark: name the failure.

The benchmark every model is graded on. The glue between labs and the companies that buy their data.

A game, not a report card.

Every batch ranked on its untreated wells, never on results.

LAB	BATCHES	MEDIAN SCORE	BEST	STREAK	MEDIAN TYPICAL	MEDIAN TIGHT	MEDIAN STEADY	MEDIAN COUNT
lab 5	33	70	#1	33	3.35	0.27	0.81	0.31
lab 6	13	60	#16	13	4.44	0.41	1.23	0.12
lab 10	17	54	#10	17	3.84	0.62	1.21	0.23
lab 8	4	52	#54	4	4.10	1.08	0.86	0.09
lab 11	5	47	#23	2	3.79	0.87	1.57	0.17
lab 3	13	44	#18	6	3.27	0.80	2.01	0.29
lab 7	8	42	#69	8	6.38	0.58	1.17	0.33
lab 2	13	39	#48	10	5.68	0.57	1.99	0.14
lab 1	6	32	#78	6	8.68	1.10	2.03	0.09
lab 9	11	25	#97	11	9.97	0.50	1.76	0.45
lab 15	6	23	#68	6	5.53	0.87	1.69	0.44

THE SCORE

Four measures in within-plate spreads, each a percentile among 129 batches: distance from the field’s centre, spread of the batch’s plates, distance from the previous batch, cells per well against the field. 100 minus the mean.

THE BEST BATCH

lab 5, 28 Jun, 7 plates: score 82. Lab 5 holds 9 of the top ten and the longest streak, 33.

YOUR CHOICE TO PLAY

Nothing in the score touches a treated well or a result, and nobody is named without opting in. The board is for climbing; the private version is for your own plates.

Control wells leave. Nothing else does.

A control well has no compound, no target and no result, so there is nothing to clear.

WHAT LEAVES THE BUILDING

untreated control wells only

images or the well-level profile of the wells with no compound. No treated well, no compound identity, no target, no result.

a plate map of the control wells

which wells are controls; nothing about what the other wells hold.

nothing else

on this public replay that is 82,319 untreated wells from 10 other labs, and not one treated well.

WHAT DATUM LINE DOES WITH IT

builds the map

one centroid per plate, in a space scaled on every other lab. The lab being placed never sets its own scale (3,773 features here).

answers

where the plate sits, distance from the lab's own baseline and to the nearest lab, cells per well, the verdict, the rank.

computes a move, not a model of you

the field's frame is one matrix per batch, fitted on your controls: $x' = C_{\text{field}}^{\frac{1}{2}} C_{\text{batch}}^{-\frac{1}{2}} (x - m_{\text{batch}})$, 200 components.

WHAT IT NEVER SEES

your treated data

the move is sent to you and applied on your side. Your results are normalised in your building, by you.

your name, unless you choose

labs appear on the map and the board as a number by default; a lab opts in to a name. Every JUMP lab here is a number.

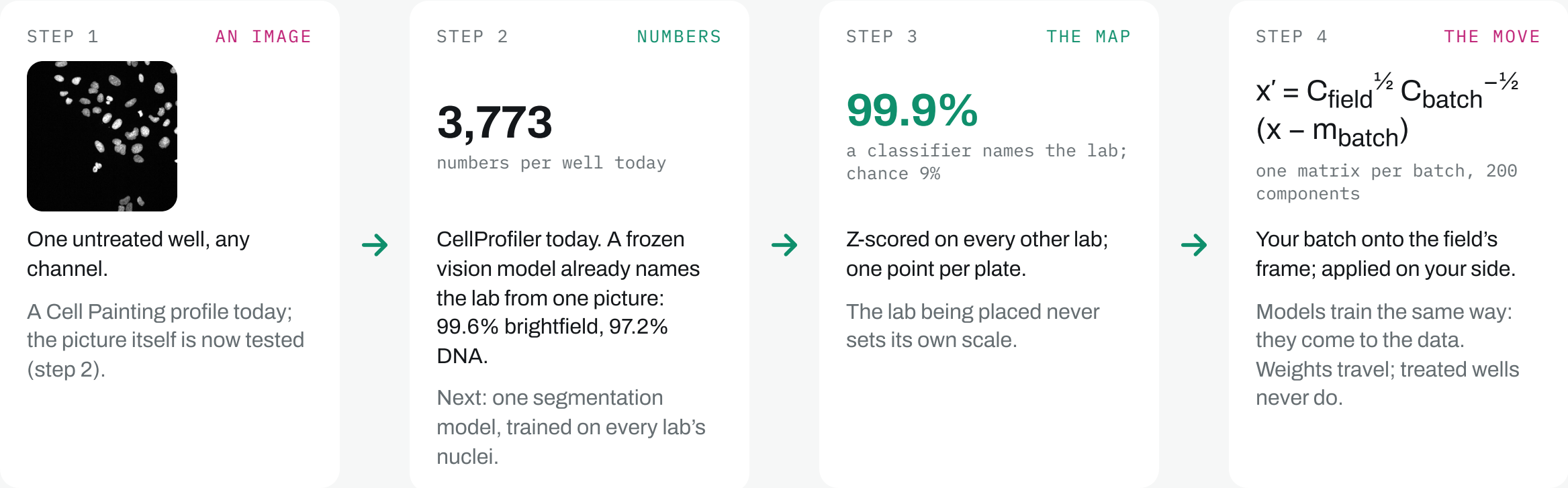
your images, for long

proposed: images are reduced to profiles on arrival and the originals deleted; the profile of a control well is what is kept.

Every analysis in this deck runs on this rule and is public: github.com/kat-titter/datum-line/analysis. The public map is JUMP, CC0 [7]. Retention and naming are proposed, not yet contracted.

Image in. Place, distance, move out.

Four steps. Data stays where it is; models and moves travel.



PEOPLE	AGENTS	BOTH
the box, the glimpse, the game	the API, the coordinates, the benchmark; models trained on every lab's controls	every drop sharpens the map; every graded model raises the bar

One laptop runs the whole replay from cache in minutes. Every script: github.com/kat-titter/datum-line/analysis. What leaves a lab: A18.

Cells cannot lie about where they came from.

Provenance, integrity and a tripwire, from the same map.

PROVENANCE

99.9%

of 93,228 untreated wells name their true lab, on plates the classifier never saw.

A plate claimed from one lab and made in another is caught by its cells. In the replay, 119 of 129 batches have every well naming its own lab (results/e1, e9).

INTEGRITY

4 batches

65 plates whose plate map says positive control, and whose cells show no effect; no other well on those plates does either.

The data said what was done to the cells, not what the label said. Left out or mislabelled, the map cannot tell; that it is wrong, it can (results/e11).

CELLS FROM NOWHERE

9 spreads

the farthest any of 129 real batches sits from every other lab. Beyond that, a culture is not the line it claims, or not alone.

Misidentified lines and contamination are the common case; untested here, and proposed as the first partner-lab test.

SENTINELS

proposed

An untreated well is a canary. If the controls move and nobody changed anything, something in the room did: a lot, a reagent, the air, the water.

Lab 2's autumn is what that looks like on public data (A13). Alerts across labs come after the certificate.

Defensive by construction: the map holds untreated wells, the benchmark grades methods on planted imaging failures, and no result or perturbation is stored. JUMP [7].

About \$1.8M for eighteen months.

People first, then the benchmark.

founder	18 months	\$180k
technical co-founder	18 months	\$180k
ML engineer	from month 7	\$160k
payroll overhead	20% of salaries	\$104k
the benchmark dataset	about 60 plates with planted failures, run where the cells already are; a CRO if needed	\$120k
partner-lab bench time	three labs image the controls they already take; reagent top-ups and shipping	\$30k
compute and storage	images, models, hosting, the API	\$45k
legal, IP, insurance, accounting	incorporation to first contract	\$45k
travel and partner visits	two benchmark partners, three labs	\$20k
as planned	the lines above	\$884k
the 2× rule	most things take two to three times what they should; plan at two	\$884k
eighteen months	about	\$1.77M

NO IN-HOUSE LAB

Partner labs image the controls they already take; the benchmark runs where the cells are, or at a CRO. A lab of our own would cost more and prove less.

THE UGLY DATASET IS INSIDE

\$150k of the \$0.9M planned, about 17%, builds the benchmark and the bench time around it. It is milestone three, not a later round.

WHAT IT HAS TO SHOW

The four milestones on slide 09. Then a seed.

Salaries are below market and assume founders are paid. The multiplier is the only line not itemised, and the one most often right. Nothing here is committed.

About \$84M a year, if everyone joined.

ASSUMPTIONS, NOT FACTS; TWO COUNTS WOULD REPLACE MOST OF THEM

Bottom up.

assumption	low	base	high
organisations screening a few lines deep	200	400	800
lines each runs per quarter	2	4	8
price per line per quarter, \$k	5	10	15
organisations screening panels of hundreds to thousands of lines	10	25	50
site licence per year, \$k	50	150	300
companies buying cell data they did not make	50	150	300
purchased datasets each certifies per year	2	4	8
price per certificate, \$k	10	25	50
benchmark partners	4	10	20
membership per partner per year, \$k	50	100	200
subscriptions and site licences, per year	\$8M	\$68M	\$399M
certificates on purchased data, per year	\$1M	\$15M	\$120M
benchmark membership, per year	\$200k	\$1M	\$4M
serviceable, per year, at full adoption	\$10M	\$84M	\$523M

TWO KINDS OF LAB

Some run a few lines deep: four, say, a primary line and a co-culture. Some run thousands of genotypes wide. Per line fits the first; a site licence fits the second. Both are rows above.

THE ANCHOR

One screen run on cells that had moved is about \$2.6M to redo. A certificate is a slice of that; a subscription is insurance against it.

TEN PERCENT

A tenth of the base case is about \$8M a year: the five-year bar.

Prices are the proposal on A15. Academic labs and cores are free and are not in the revenue rows; they are the map.